EEG Dataset Analysis & Classification

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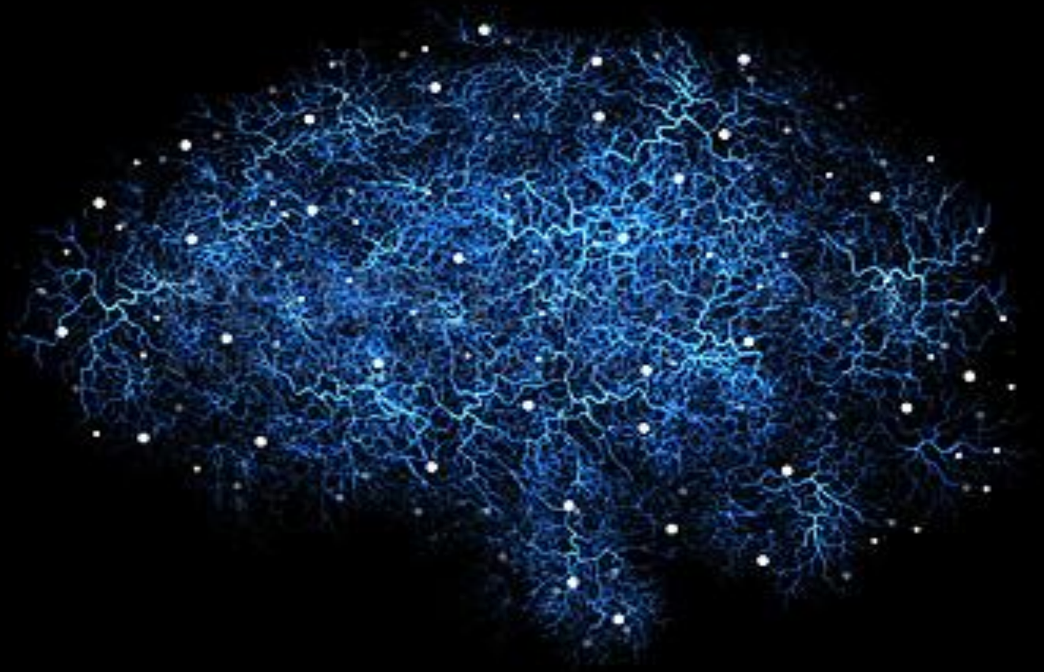
- Spring 1404 -

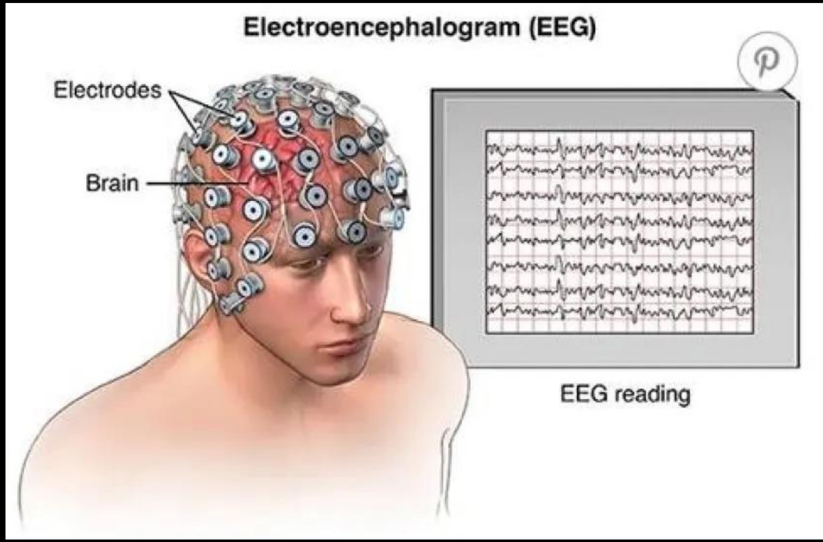
Agenda

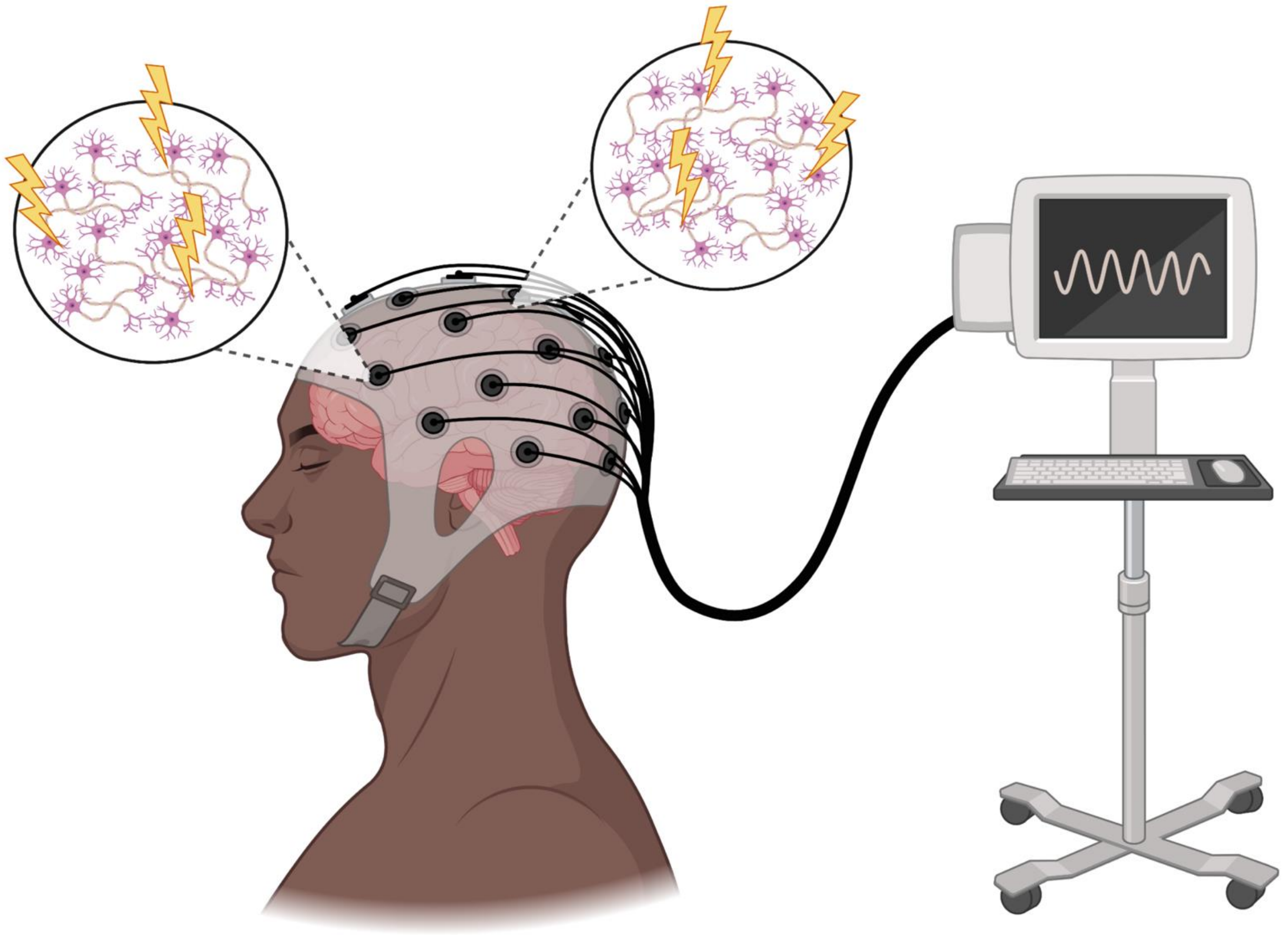
- What’s EEG?
- Why EEG classification?
- Our Dataset
- Preprocessing Steps
- Power BI EDA
- Model Architecture
- Training & Evaluation
- Inference Examples
- Key Findings & Next Steps

What’s EEG?

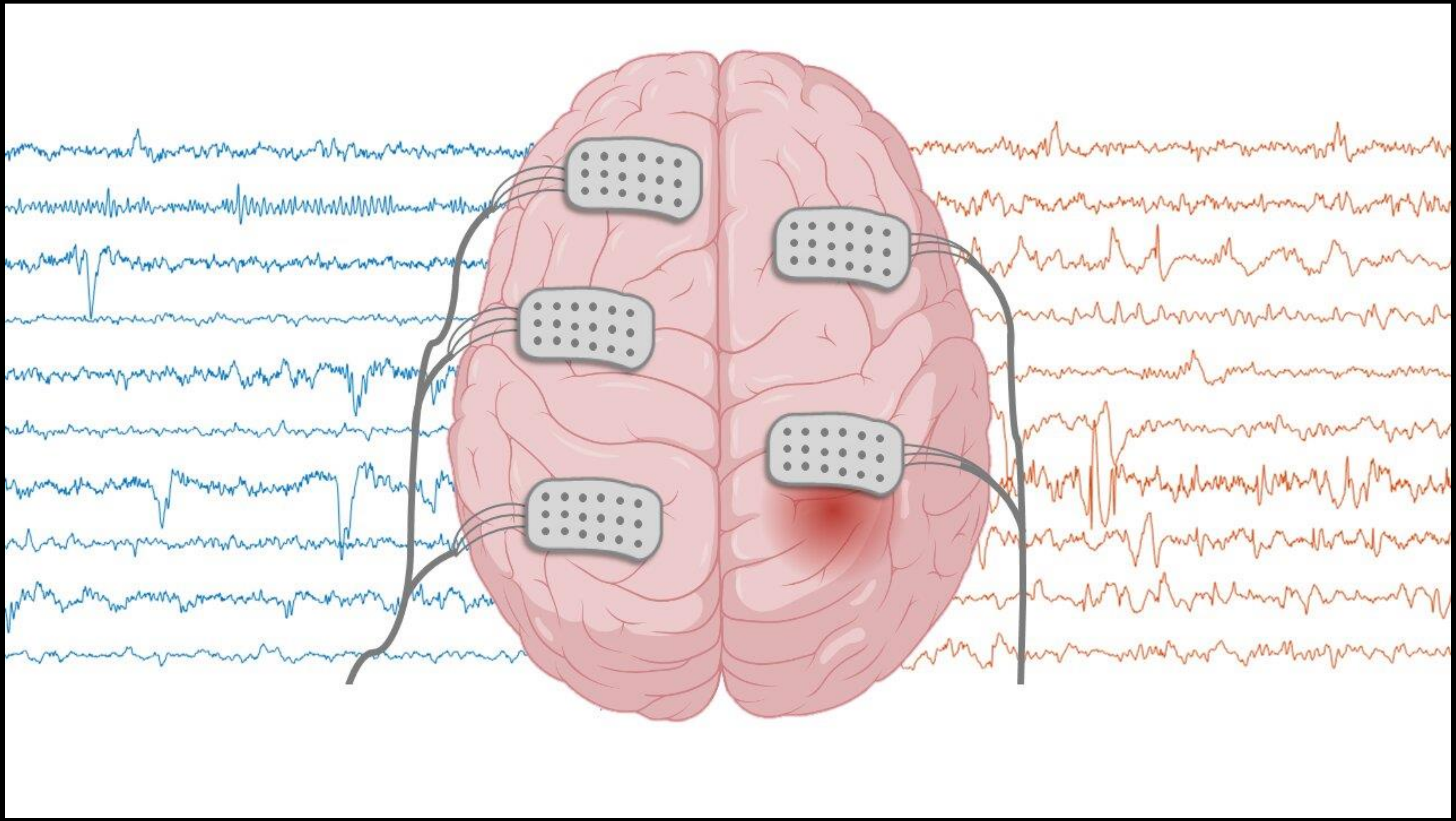


- EEG (Electroencephalography) records brain activity using electrodes
- Used in neuroscience, medical diagnosis, and brain-computer interfaces
- Motor Imagery (MI): Imagining movement without actual motion
- Easy language: “We’re trying to read imagined hand movement from brain signals.”





Why EEG Classification?



- Brain-computer interfaces (BCIs) for paralyzed patients
- Real-time control in neuro-rehab
- Deep learning is good at decoding complex brain signals
- **Fun fact:** We’re not predicting thoughts, just left vs right imagined movement

Our Dataset

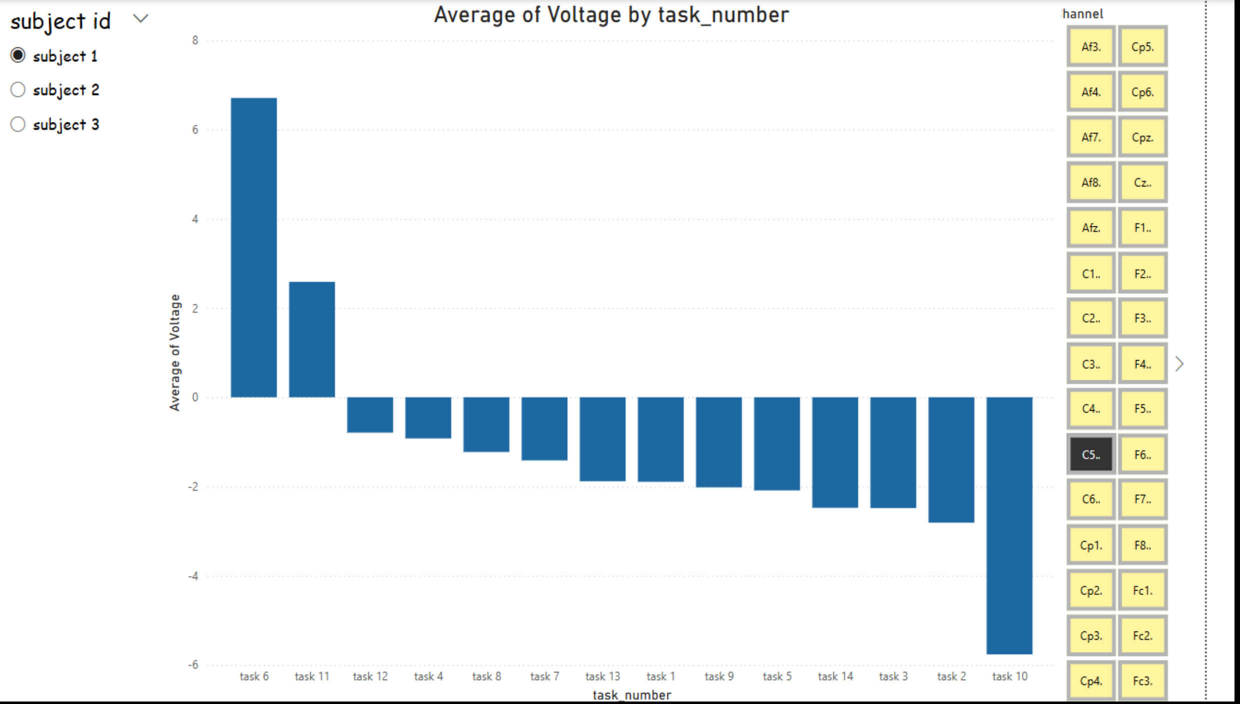
- Public EEG Motor Imagery dataset from PhysioNet
- 109 subjects, 64 EEG channels
- We used **subjects 1–79**, runs 4, 8, and 12 (imagined left/right fist)
- Each sample: 3-second segment after stimulus (called **epochs**)
- Total samples: 3377 epochs

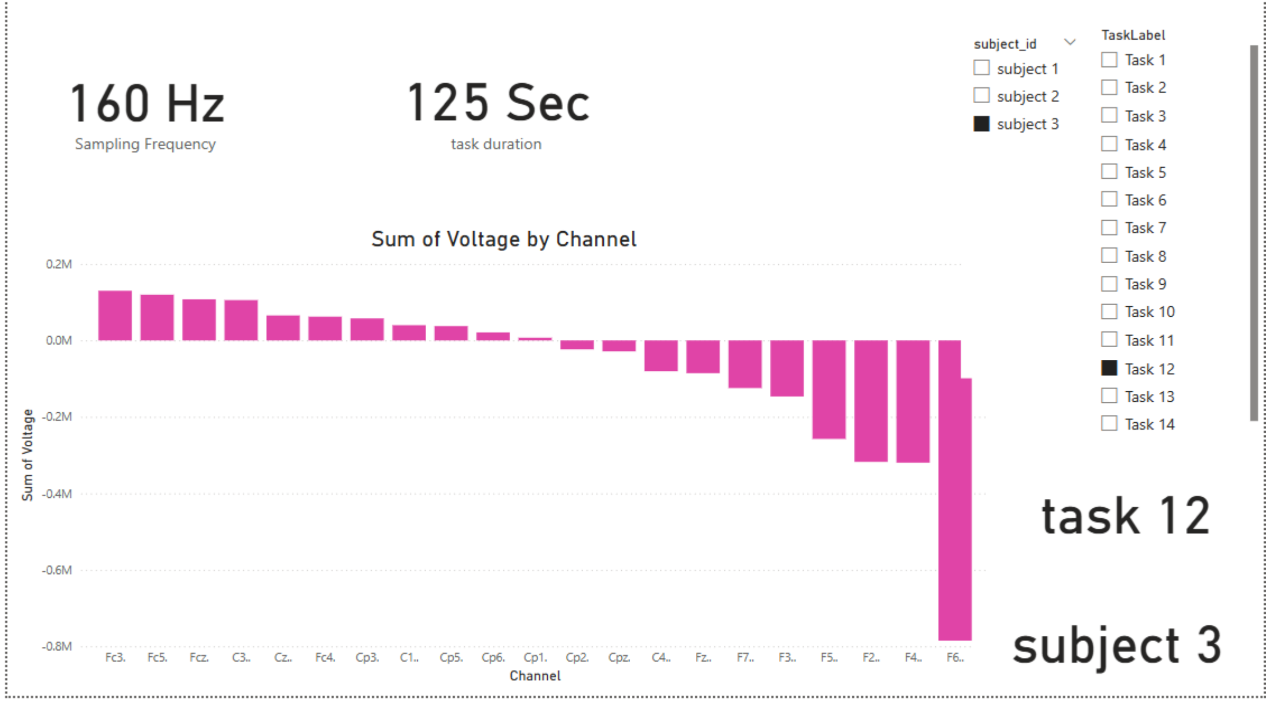
Preprocessing Steps

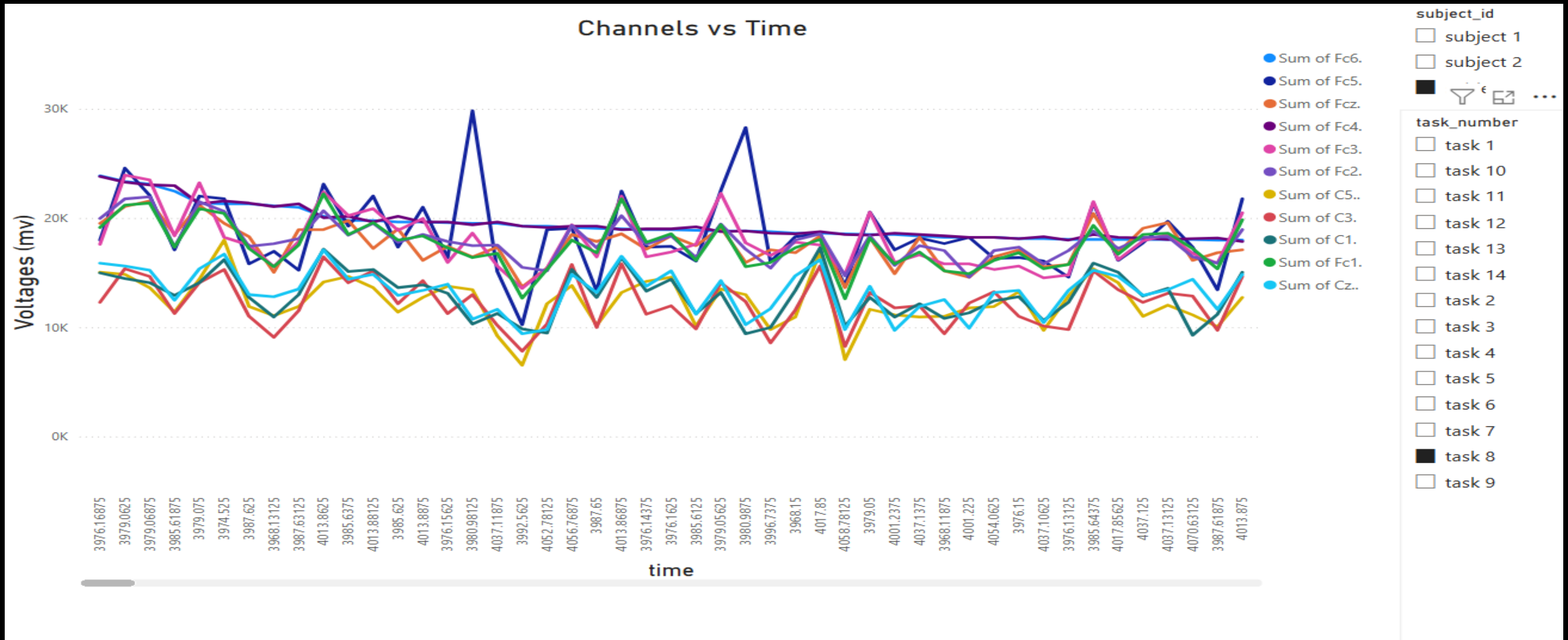
- Loaded and combined .edf files using mne
- Extracted event codes: T1 = left, T2 = right
- Epoched signals from 1s to 4.1s post-stimulus
- Converted units (Volts → millivolts)
- Data split: 81% train, 9% validation, 10% test

Power BI Visualizations

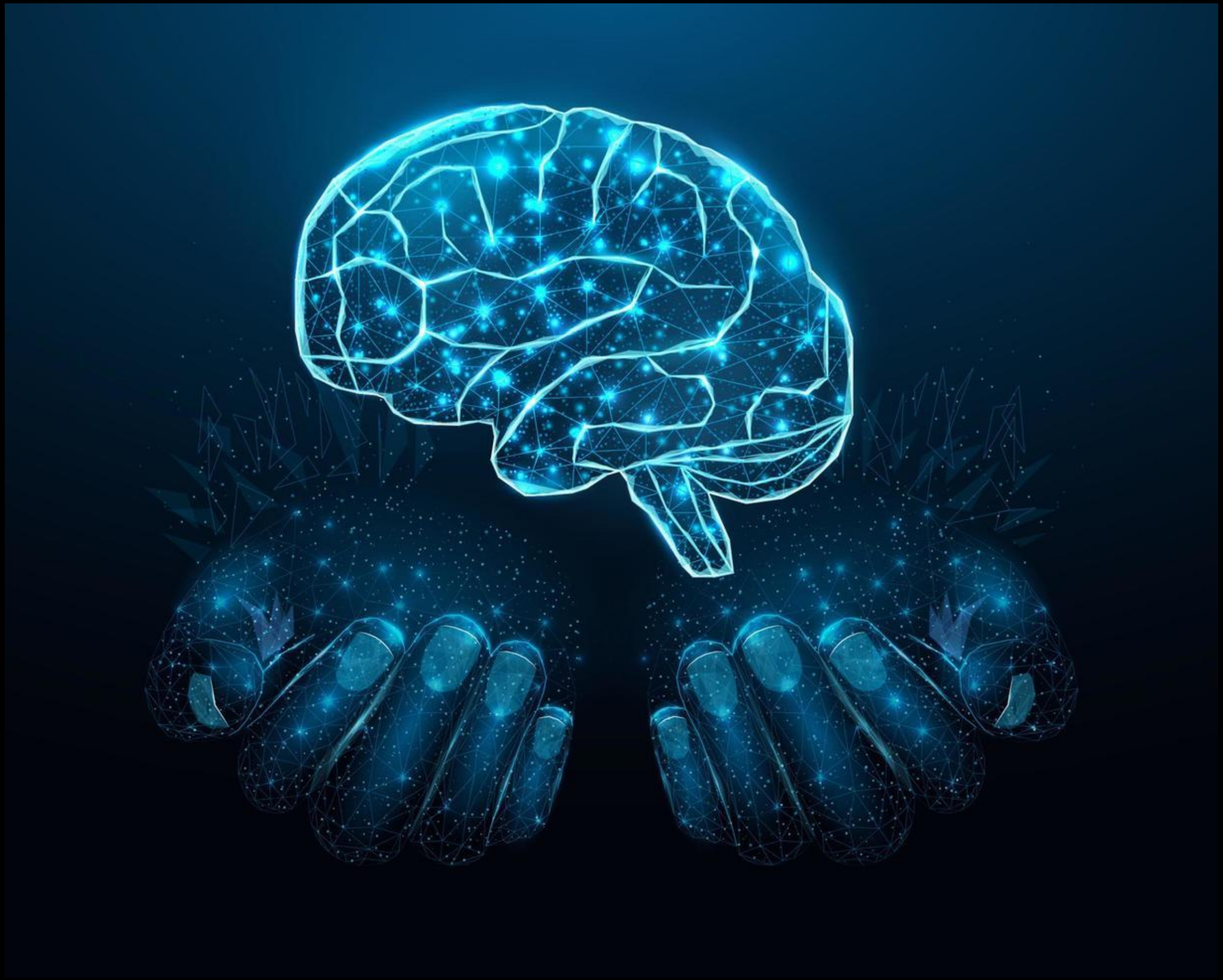
- Class distribution of left/right across subjects
- Channel-wise signal comparison
- Variance heatmaps of EEG channels
- Highlight: which channels are more active for left vs right imagery


subject id
subject 1
subject 2
subject 3
Average of Voltage by task_number
Average of Voltage
8
6
4
2
0
-2
-4
-6
task 6
task 11
task 12
task 4
task 8
task 7
task 13
task 1
task 9
task 5
task 14
task 3
task 2
task 10
task_number
hannel
Af3.
Cp5.
Af4.
Cp6.
Af7.
Cpz.
Af8.
Cz..
Afz.
F1..
C1..
F2..
C2..
F3..
C3..
F4..
C4..
F5..
C5..
F6..
C6..
F7..
Cp1.
F8..
Cp2.
Fc1.
Cp3.
Fc2.
Cp4.
Fc3.


160 Hz
Sampling Frequency
125 Sec
task duration
subject_id
subject 1
subject 2
subject 3
TaskLabel
Task 1
Task 2
Task 3
Task 4
Task 5
Task 6
Task 7
Task 8
Task 9
Task 10
Task 11
Task 12
Task 13
Task 14
Sum of Voltage by Channel
Sum of Voltage
0.2M
0.0M
-0.2M
-0.4M
-0.6M
-0.8M
Fc3.
Fc5.
Fcz.
C3..
Cz..
Fc4.
Cp3.
C1..
Cp5.
Cp6.
Cp1.
Cp2.
Cpz.
C4..
Fz..
F7..
F3..
F5..
F2..
F4..
F6..
Channel
task 12
subject 3

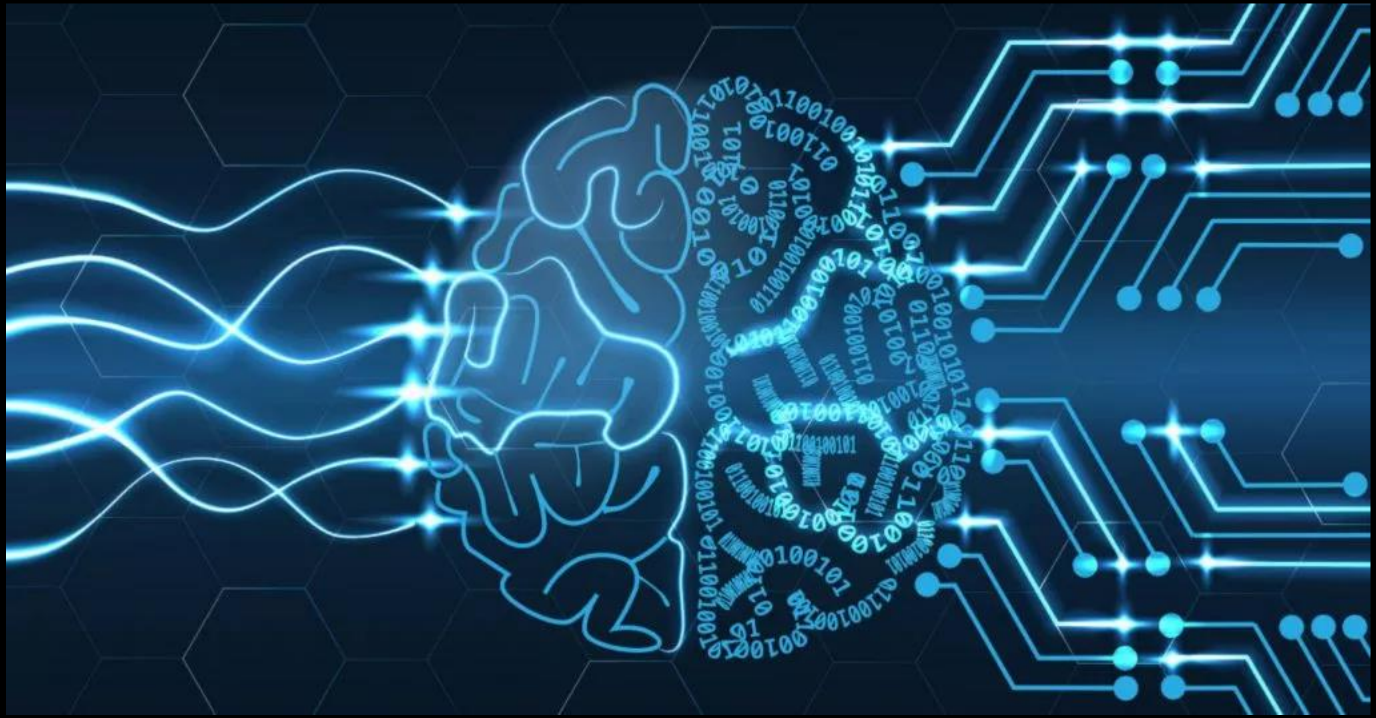

Channels vs Time
Voltages (mv)
30K
20K
10K
0K
Sum of Fc6.
Sum of Fc5.
Sum of Fcz.
Sum of Fc4.
Sum of Fc3.
Sum of Fc2.
Sum of C5..
Sum of C3..
Sum of C1..
Sum of Fc1.
Sum of Cz..
time
subject_id
subject 1
subject 2
task_number
task 1
task 10
task 11
task 12
task 13
task 14
task 2
task 3
task 4
task 5
task 6
task 7
task 8
task 9

Model Architecture

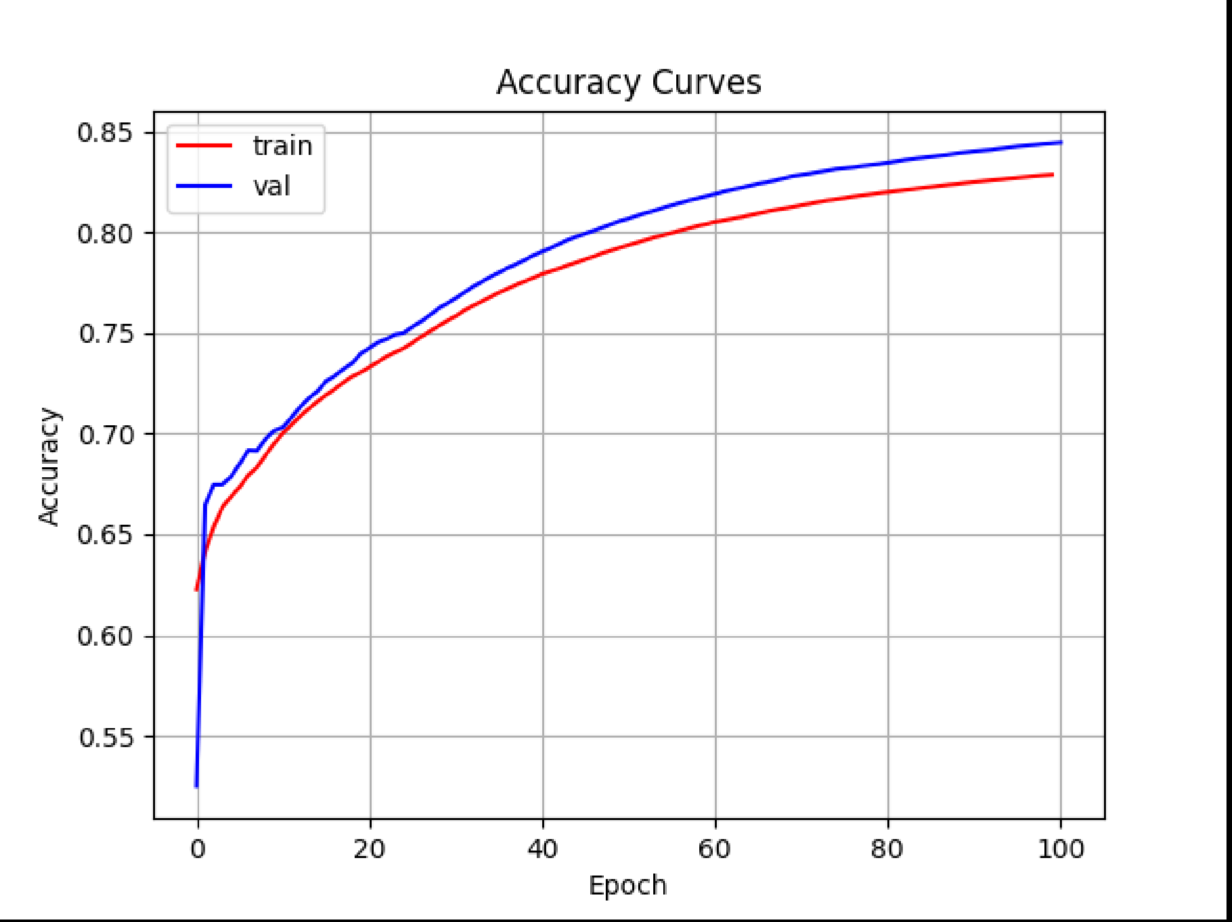


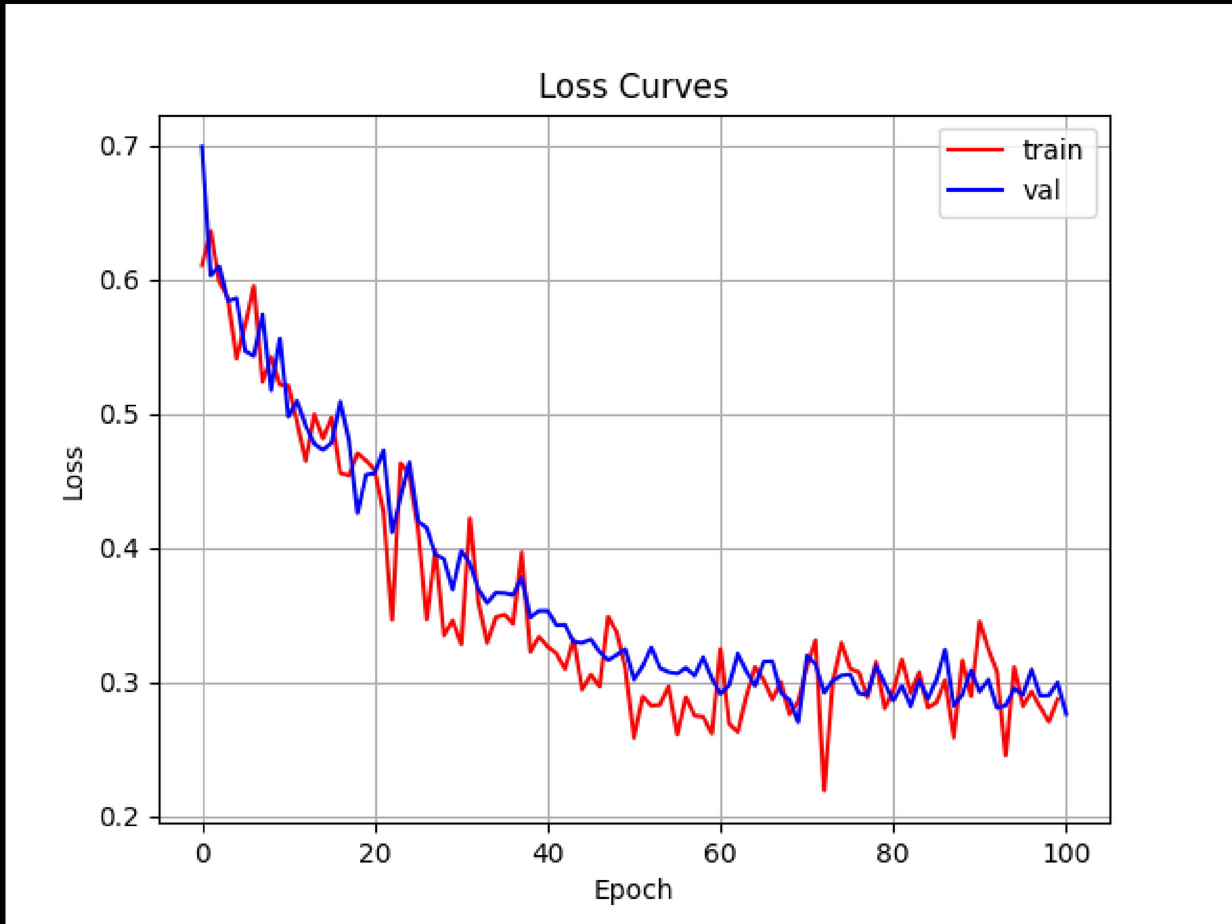
- Convolutional layer (extracts features per channel)
- Transformer layers (capture temporal patterns)
- Fully-connected layers (classify into left/right)
- Used PyTorch Lightning for modularity and training management

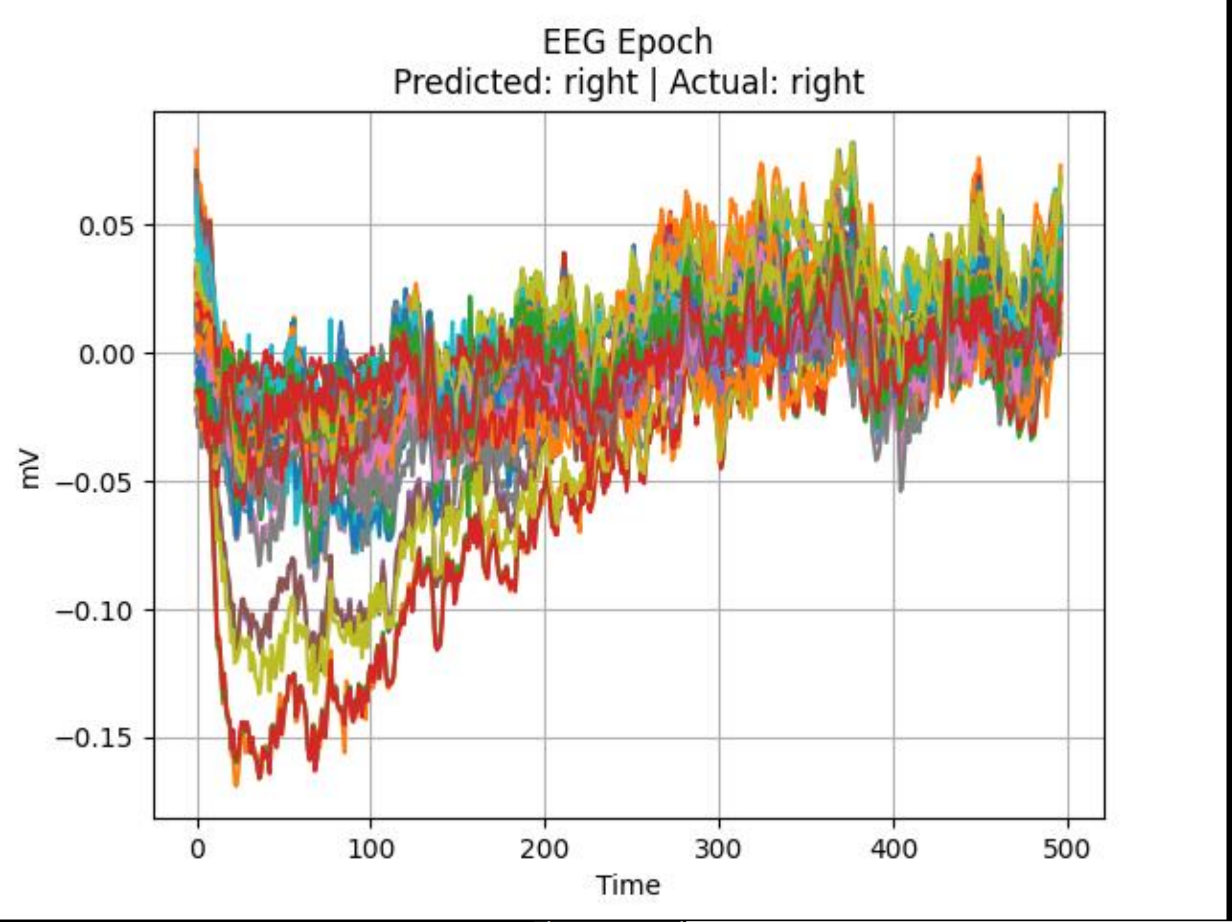
Training and Evaluation



- Trained for 100 epochs with early stopping
- Best model saved as EEGClassificationModel_best.ckpt
- Metrics:
- Train accuracy: 76%
- Validation accuracy: 84.6%
- Test accuracy: 82.4%
- Plots: Loss & Accuracy vs Epoch


Accuracy Curves
train
val
Accuracy
Epoch
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0
20
40
60
80
100


Loss Curves
train
val
Loss
Epoch
0.7
0.6
0.5
0.4
0.3
0.2
0
20
40
60
80
100



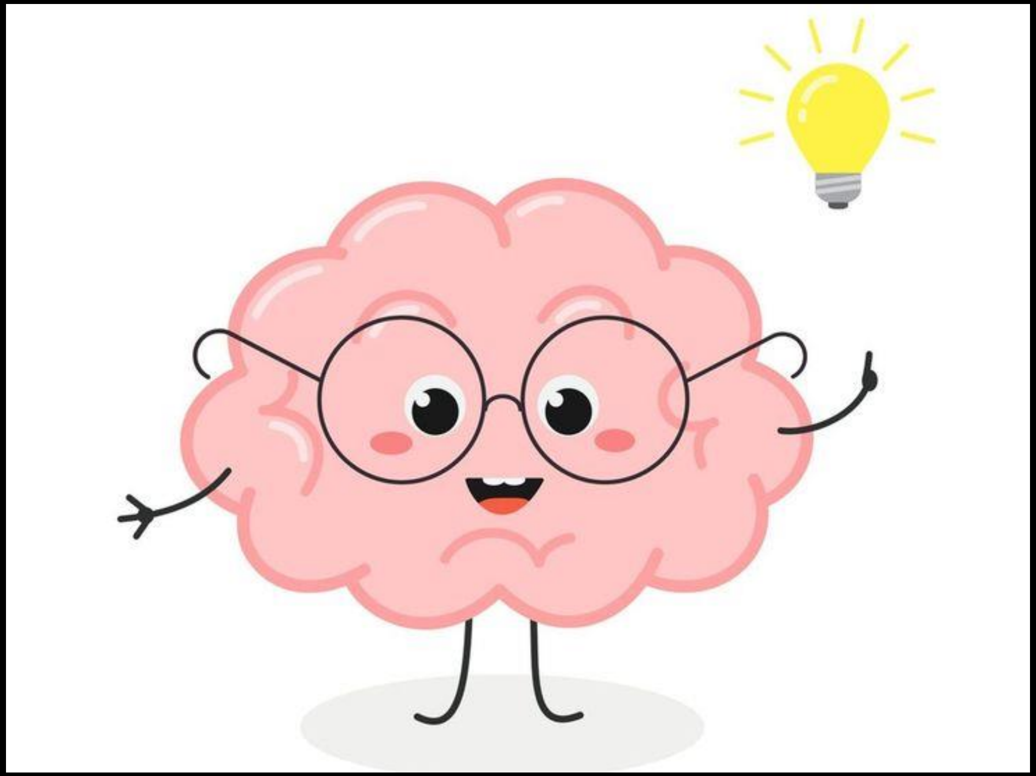
```
Trainer.fit` stopped: `max_epochs=100` reached.
Epoch 99: 100%|████████████████████████████████████| 274/274 [02:01<00:00,  2.26it/s, v_num=0_1, train_loss=0.127, train_acc=0.828, val_loss=0.276, val_acc=0.844]
Restoring states from the checkpoint path at D:\Term 8\data science\final project\DSphase3\best_model.ckpt
Loaded model weights from the checkpoint at D:\Term 8\data science\final project\DSphase3\best_model.ckpt
C:\Users\mahdi\AppData\Local\Programs\Python\Python311\Lib\site-packages\lightning\pytorch\trainer\connectors\data_connector.py:425: The 'test_dataloader' does not have many workers which may be a bott
leneck. Consider increasing the value of the `num_workers` argument` to `num_workers=7` in the `DataLoader` to improve performance.
Testing DataLoader 0: 100%|████████████████████████████████████| 338/338 [00:04<00:00, 72.20it/s]
```

Test metric	DataLoader 0
test_acc	0.7636850476264954
test_loss	0.8715420961380005

```
Restoring states from the checkpoint path at EEGClassificationModel_best.ckpt
Loaded model weights from the checkpoint at EEGClassificationModel_best.ckpt
C:\Users\mahdi\AppData\Local\Programs\Python\Python311\Lib\site-packages\lightning\pytorch\trainer\connectors\data_connector.py:425: The 'predict_dataloader' does not have many workers which may be a b
ottleneck. Consider increasing the value of the `num_workers` argument` to `num_workers=7` in the `DataLoader` to improve performance.
Predicting DataLoader 0: 100%|████████████████████████████████████| 1/1 [00:00<00:00, 76.92it/s]

Prediction: right
Actual:     right
```

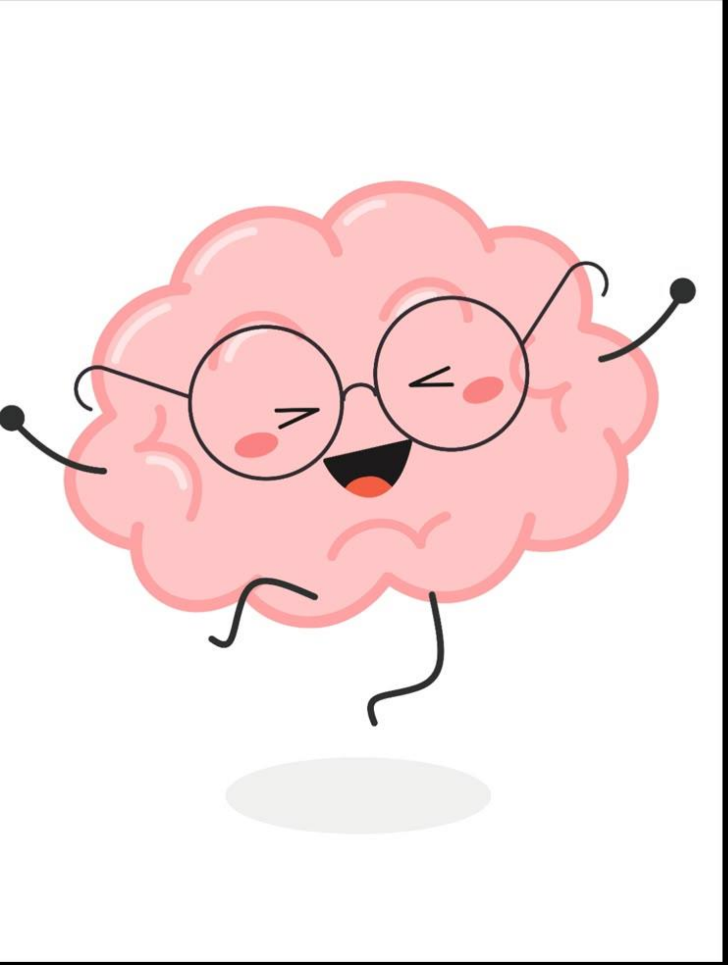
Key Findings



- EEG signals are noisy and complex, but with the right architecture, we can extract meaningful information.
- The Transformer component helped significantly in learning temporal dependencies.
- Power BI visualizations allowed us to identify the most informative EEG channels early.
- The model showed stable and reliable performance, especially thanks to a modular and automated pipeline.

Next Steps

- Add more tasks (like foot vs hand vs rest)
- Try CNN-LSTM or other architectures
- Real-time BCI implementation (future project)
- Investigate signal quality by channel or subject



Thank You!